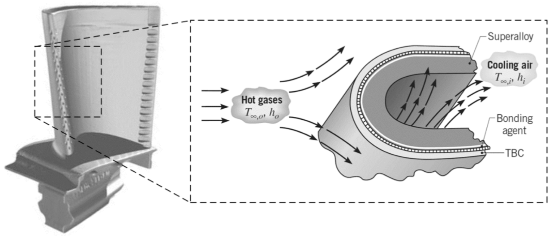
3. (30 pt) Energy efficiency of a gas turbine can be improved by operating it under higher temperature. However, hot gases produced by the combustor pose a significant challenge to the reliable operation of turbine blades. An approach to protecting the turbine blades from material failure in the high-temperature environment is to apply a "thermal barrier coating (TBC)" to the exterior surface of the blade, as shown in the figure below. The blade is made from a high-temperature superalloy with thermal conductivity of k_b = 25 W/(m·K). The TBC is made of a ceramic material with thermal conductivity of k_{TBC} = 1 W/(m·K). To attach the TBC to the exterior surface of the superalloy, a metallic bonding agent (thickness negligible) is needed, which creates an interfacial thermal resistance (per unit area) of R'' = 10^{-4} m^2·K/W. Consider an operating condition for which external hot gases at $T_{\infty,o}$ = 1616 K and cooling air inside the blade at $T_{\infty,i}$ = 400 K. Heat transfer coefficients of hot gases and cooling air are h_o = 1000 W/(m^2·K) and 500 W/(m^2·K), respectively. The turbine blade is at a steady-state condition, where thermal radiation can be neglected. The turbine blade can be approximated as a plane wall, so that we can assume one-dimensional heat conduction through the TBC and the superalloy. The thickness of superalloy blade is L_b = 5 mm. The maximum allowable temperature for the superalloy is T_{max} = 1200 K. Consider the following two conditions.



(a) Condition 1: the TBC is not applied. If we do not apply the TBC and directly expose the superalloy blade to hot gases, draw the thermal resistance network for this condition and label all thermal resistances. Calculate the exterior surface temperature of the superalloy blade. Can the superalloy blade be maintained below T_{max}?

(b) Condition 2: the TBC is applied. Now we apply L_{TBC} = 0.5 mm thick TBC to the superalloy blade. Draw the thermal resistance network for this condition and label all thermal resistances. Calculate the temperature of the superalloy blade again. Can the superalloy blade be maintained below T_{max} this time?

Given: $k_b = 25\ W/(mK)$, $k_{TBC} = 1\ W/(mK)$, $R''_{bonding} = 10^{-4}\ m^2 \cdot K/W$, $T_{\infty,o} = 1616\ K$, $T_{\infty,i} = 400\ K$, $h_o = 1000\ W/(m^2K)$, $h_i = 500\ W/(m^2K)$, $L_b = 5\ mm$, $T_{max} = 1200\ K$, $L_{TBC} = 0.5\ mm$

Find: Thermal resistance network and T_b when TBC not applied (part a) and when TBC is applied (part b). $T_b < T_{max}$ in either condition?

Assumptions: thickness of metallic bonding agent negligible, turbine blade at steady state, thermal radiation neglected, 1D heat conduction through TBC and superalloy

Analysis: a.

$$T_{\infty,o} \circ\!\!-\!\!\underset{}{\overset{R''_{conv,o}}{\sim\!\sim}}\!\!-\!\!\underset{T_{b,o}}{\circ}\!\!-\!\!\overset{R''_b}{\sim\!\sim}\!\!-\!\!\underset{T_{b,i}}{\circ}\!\!-\!\!\overset{R''_{conv,i}}{\sim\!\sim}\!\!-\!\!\circ\ T_{\infty,i}$$

$$R''_{conv,o} = \frac{1}{h_o} = \frac{1}{1000\ W/(m^2K)} = 0.001\ m^2K/W$$

$$R''_b = \frac{L_b}{k_b} = \frac{0.005\ m}{25\ W/(mK)} = 2 \times 10^{-4}\ m^2K/W$$

$$R''_{conv,i} = \frac{1}{h_i} = \frac{1}{500\ W/(m^2K)} = 0.002\ m^2K/W$$

$$R''_{tot} = R''_{conv,o} + R''_b + R''_{conv,i} = 0.001\ m^2K/W + 2 \times 10^{-4}\ m^2K/W + 0.002\ m^2K/W = 0.0032\ m^2K/W$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616K - 400K}{0.0032\ m^2K/W} = 380000\ W/m^2$$

Problem 3 continued

(a.) $T_{\infty,o} - T_{b,o} = q'' R''_{conv,o}$

$T_{b,o} = T_{\infty,o} - q'' R''_{conv,o} = 1616\,K - 380000\,W/m^2 \times 0.001\,m^2K/W = 1236\,K = T_{b,o}$

$1236\,K > 1200\,K$, $\therefore$ the superalloy blade cannot be maintained below T_{max}

b.

$$T_{\infty,o} \circ\!\!\xrightarrow{R''_{conv,o}}\!\!\circ_{T_{TBC,o}} \xrightarrow{R''_{TBC}} \circ_{T_{TBC,i}} \xrightarrow{R''_{bonding}} \circ_{T_{b,o}} \xrightarrow{R''_{b}} \circ_{T_{b,i}} \xrightarrow{R''_{conv,i}} \circ\ T_{\infty,i}$$

$$R''_{TBC} = \frac{L_{TBC}}{k_{TBC}} = \frac{0.0005\,m}{1\,W/(mK)} = 5\times10^{-4}\ m^2K/W$$

$$R''_{tot} = 0.0032\,m^2K/W + 5\times10^{-4}\,m^2K/W + 10^{-4}\,m^2K/W = 0.0038\,m^2K/W$$

$$q'' = \frac{1616\,K - 400\,K}{0.0038\,m^2K/W} = 320000\ W/m^2$$

$T_{TBC,o} = 1616\,K - 320000\,W/m^2 \times 0.001\,m^2K/W = 1296\,K$

$T_{TBC,o} - T_{TBC,i} = q'' R''_{TBC}$

$T_{TBC,i} = T_{TBC,o} - q'' R''_{TBC} = 1296\,K - 320000\,W/m^2 \times 5\times10^{-4}\,m^2K/W = 1136\,K$

$T_{b,o} = 1136\,K - 320000\,W/m^2 \times 10^{-4}\,m^2K/W = 1104\,K = T_{b,o}$

$1104\,K < 1200\,K$ $\therefore$ the superalloy blade can be maintained below T_{max}